

JORDAN A. STERLING

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PROFESSIONAL SUMMARY

Highly accomplished Senior Backend Developer with over 10 years of experience in architecting, developing, and deploying mission-critical distributed systems and cloud-native applications. Expert in steering high-level technical strategy by aligning engineering roadmaps with overarching business objectives to ensure scalable growth and operational excellence across multi-tenant SaaS environments. Proven track record of leading cross-functional teams through the full software development life cycle, emphasizing robust API design, high-concurrency processing, and rigorous automated testing protocols. Strategic visionary focused on optimizing system performance through sophisticated caching mechanisms, database sharding, and asynchronous message queuing to support millions of concurrent users. Committed to fostering a culture of continuous improvement, mentoring junior engineers in clean code principles, and implementing advanced CI/CD pipelines to accelerate deployment frequency and system reliability.

PROFESSIONAL EXPERIENCE

NEXGEN FINANCIAL SOLUTIONS | LEAD BACKEND ENGINEER

January 2020 - Present

- Spearheaded the architectural redesign of the core payment processing engine from a monolithic structure to a Go-based microservices architecture, effectively increasing transaction throughput by 300% and reducing latency by 150ms per request.
- Engineered a custom distributed locking mechanism using Redis and Redlock to prevent double-spending and ensure transactional integrity across globally distributed data centers during high-volume trading windows.

- Orchestrated the migration of legacy on-premise relational databases to a horizontally scalable Amazon Aurora cluster, implementing a multi-region failover strategy that achieved 99.99% uptime for the primary financial ledger.
- Developed a comprehensive GraphQL API layer to unify disparate data sources for the frontend team, reducing the number of network round-trips by 60% and significantly improving the mobile application's perceived performance.
- Implemented an advanced observability stack utilizing Prometheus, Grafana, and Jaeger to provide deep-link tracing across 50+ microservices, allowing the team to reduce Mean Time to Resolution (MTTR) by 45%.
- Led a team of 12 backend developers in adopting Test-Driven Development (TDD) and Behavior-Driven Development (BDD), resulting in a 35% decrease in production bugs and a significant increase in code maintainability.
- Designed a real-time fraud detection pipeline using Apache Kafka and Flink to analyze transaction patterns in sub-seconds, successfully identifying and blocking over \$2M in unauthorized transactions within the first quarter.
- Automated the end-to-end deployment process using Terraform and Kubernetes, enabling blue-green deployment strategies that eliminated downtime during version rollouts and simplified the rollback process.
- Formulated a comprehensive API versioning and deprecation strategy to support legacy enterprise clients while allowing the product team to iterate rapidly on new features without breaking backward compatibility.
- Mentored mid-level engineers through structured pair programming sessions and technical design reviews, facilitating the promotion of four team members to senior-level roles within an 18-month period.

GLOBAL RETAIL SYSTEMS | SENIOR SOFTWARE ENGINEER

June 2016 – December 2019

- Architected a high-concurrency inventory management system using Java and Spring Boot that handled over 50,000 requests per second during peak seasonal sales events without any degradation in system performance.
- Optimized complex SQL queries and implemented a multi-tiered caching strategy involving Caffeine (L1) and Memcached (L2), which reduced database load by 70% and cut page load times for the product catalog by 1.2 seconds.

- Integrated third-party logistics and shipping APIs (FedEx, UPS, DHL) into a centralized fulfillment gateway, utilizing the Adapter pattern to standardize data formats and simplify the addition of new carriers.
- Developed a robust event-driven architecture using RabbitMQ to decouple order placement from downstream services such as invoicing, notification, and shipping, ensuring system resilience during service outages.
- Led the initiative to containerize the entire backend ecosystem using Docker, providing developers with consistent local environments and reducing "it works on my machine" issues by approximately 80%.
- Established a rigorous code review process focused on security best practices, specifically targeting OWASP Top 10 vulnerabilities, which led to a flawless record in external security audits over a three-year period.
- Designed and implemented a self-service internal developer portal that automated the scaffolding of new microservices, reducing the time from project conception to first deployment from two weeks to under two hours.
- Collaborated with Data Science teams to build a high-throughput data ingestion pipeline in Python, feeding real-time user behavior data into a Spark-based recommendation engine to increase cross-sell conversion by 12%.
- Revamped the authentication and authorization system using OAuth2 and OpenID Connect with Okta integration, providing secure and seamless Single Sign-On (SSO) capabilities for both internal staff and external partners.

CLOUDSTREAM TECHNOLOGIES | BACKEND DEVELOPER

July 2013 ? May 2016

- Developed and maintained RESTful web services using Python and Django for a video streaming platform, focusing on efficient metadata retrieval and user preference management for a global audience.
- Built a scalable content ingestion system that utilized AWS S3 and Lambda to automatically transcode uploaded video files into multiple bitrates and formats, optimizing delivery for various device types and network conditions.
- Managed a large-scale migration from a self-hosted MongoDB cluster to a managed MongoDB Atlas environment, improving data availability and automating routine maintenance tasks like backups and patching.

- Created a comprehensive suite of unit and integration tests using PyTest, achieving over 85% code coverage and ensuring the stability of the core business logic during rapid feature expansion phases.
- Optimized the search functionality of the platform by implementing Elasticsearch with custom analyzers and scoring functions, resulting in a 25% improvement in search relevance and user engagement metrics.
- Built internal monitoring dashboards using ELK Stack (Elasticsearch, Logstash, Kibana) to visualize system health and API usage patterns, enabling proactive capacity planning for upcoming marketing campaigns.
- Collaborated with Frontend developers to implement WebSockets for real-time chat and live-streaming interaction features, supporting concurrent connections for over 100,000 viewers during live events.
- Documented all internal and external APIs using Swagger/OpenAPI specifications, providing clear and interactive documentation that reduced the onboarding time for new developers and external integrators.

TECHNICAL PROJECTS

DISTRIBUTED LEDGER AND RECONCILIATION ENGINE

- Developed a high-performance reconciliation engine in Go designed to match millions of financial records per minute between internal databases and external bank statements.
- Utilized an actor-based concurrency model to process data streams in parallel, ensuring that the system remained responsive even under heavy computational load.
- Implemented a custom hashing algorithm to verify data integrity across disparate storage systems, identifying discrepancies caused by network jitter or hardware failure with 100% accuracy.
- Integrated a reporting module that generated detailed audit trails and compliance reports, meeting strict regulatory requirements for financial transparency and data governance.

MULTI-TENANT SAAS ORCHESTRATOR

- Engineered a centralized management platform to handle tenant provisioning, configuration, and resource isolation for a large-scale B2B SaaS application.
- Developed a dynamic routing layer using NGINX and Lua to direct traffic to tenant-specific shards based on headers, ensuring low latency and strict data privacy.
- Created a robust billing integration with Stripe that dynamically adjusted subscription tiers based on real-time resource consumption and user seat counts.
- Automated the lifecycle of tenant databases using Liquibase for schema migrations, ensuring that all 500+ tenants were running synchronized versions of the application schema.

REAL-TIME ANALYTICS AND STREAMING PIPELINE

- Built a data processing pipeline using Apache Pulsar and ClickHouse to ingest and analyze billions of telemetry events from IoT devices in real-time.
- Designed a custom serialization protocol using Protocol Buffers (Protobuf) to minimize the network footprint of telemetry data, resulting in a 40% reduction in bandwidth costs.
- Implemented complex windowing functions to calculate moving averages and anomaly detection scores, triggering automated alerts when device metrics exceeded predefined thresholds.
- Developed a high-performance visualization API that allowed users to query multi-dimensional datasets with sub-second response times, even across petabytes of historical data.

AUTONOMOUS DEPLOYMENT AND HEALER SYSTEM

- Created a custom internal tool in Python that monitors Kubernetes cluster health and automatically triggers rollbacks or resource scaling based on predictive AI models.
- Integrated with the Slack API to provide real-time deployment notifications and interactive buttons for manual intervention during critical production incidents.
- Implemented a "chaos engineering" component that periodically injected failures into the staging environment to validate the system's self-healing capabilities and resilience.

- Reduced the average downtime caused by failed deployments by 90% through the use of automated health checks and canary analysis.

MICROSERVICES API GATEWAY TRANSFORMATION

- Led the implementation of an Enterprise API Gateway using Kong to centralize cross-cutting concerns such as rate limiting, logging, and request transformation.
- Developed custom plugins in Lua to handle proprietary authentication headers and perform real-time request validation against JSON schemas.
- Optimized the gateway's performance to handle 100k+ RPS with sub-5ms overhead, ensuring that the introduction of the gateway did not negatively impact end-user experience.
- Implemented a service discovery mechanism using Consul to allow the gateway to dynamically route traffic to healthy service instances in a constantly changing cloud environment.

TECHNICAL SKILLS

LANGUAGES: Go (Golang), Java, Python, C++, Scala, Ruby, JavaScript (Node.js), SQL, Lua, Rust.

BACKEND FRAMEWORKS: Spring Boot, Django, Flask, Express.js, Gin, Echo, Rails, Micronaut, Fastapi.

DATABASES & STORAGE: PostgreSQL, MySQL, Amazon Aurora, MongoDB, Cassandra, Redis, Memcached, Elasticsearch, ClickHouse, DynamoDB, Oracle.

INFRASTRUCTURE & DEVOPS: AWS (EC2, S3, RDS, Lambda, ECS, EKS), Google Cloud Platform (GCP), Kubernetes, Docker, Terraform, Helm, Ansible, Jenkins, GitLab CI, GitHub Actions.

MESSAGING & STREAMING: Apache Kafka, RabbitMQ, Apache Pulsar, Amazon SQS/SNS, Apache Flink, Spark Streaming.

TESTING & QUALITY: JUnit, PyTest, Testify, Mockito, Selenium, Postman, SonarQube, Locust (Load Testing), JMeter.

NETWORKING & SECURITY: gRPC, Protobuf, REST, GraphQL, OAuth2, OpenID Connect, JWT, TLS/SSL, VPC, IAM, WAF.

TOOLS & METHODOLOGIES: Git, JIRA, Confluence, Agile/Scrum, Kanban, Domain-Driven Design (DDD), Microservices, Serverless Architecture, Twelve-Factor App.

EDUCATION

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

University of Technology, Austin, TX

CERTIFICATIONS

- AWS Certified Solutions Architect ? Professional
- Certified Kubernetes Administrator (CKA)
- Oracle Certified Professional, Java SE Programmer
- Google Professional Cloud Developer